SINGAPORE UNIVERSITY OF SOCIAL SCIENCES

MTH357

Examination - January Semester 2024

Applied Regression Analysis I

Thursday, 18 April 2024 **04:00 pm - 06:00 pm**

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FOUR (4)** questions and comprises **EIGHT (8)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is an **Open-book** examination.

At the end of the examination.
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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(Full marks: 100)

Question 1

A gardener is keen to predict the height of sunflowers in his garden based on the number of days since he planted the seeds. He meticulously documents the height of 5 sunflower stalks at 9 different time points for a total of 45 observations.

The variables of interest are as follows:

y: Height of the sunflower (in centimeters).

x: Number of days since planting (in days).

With this data, a Simple Linear Regression (SLR) model has been applied to estimate the relationship between these variables:

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \varepsilon_i \sim N(0, \sigma^2).$$

Here are some of the key summary statistics derived from the analysis:

$$\sum_i x_i y_i = 70,050$$

$$\sum_i x_i^2 = 35,625$$

$$\sum_i y_i^2 = 140,621.1$$

$$\overline{x} = 25$$

$$\overline{y} = 49.23503$$

Question 1a

Estimate the coefficients β_0 and β_1 and interpret the point estimates.

(6 marks)

Question 1b

An R output from summary() and lm() is given below.

```
> summary(lm1)

Call:
lm(formula = height ~ days, data = sunflower)

Residuals:
    Min      1Q  Median      3Q     Max 
-16.453  -4.346  -0.008   3.135  14.799 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  [redacted]  2.65861  [redacted]
days         [redacted]  0.09449  [redacted]
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 8.183 on 43 degrees of freedom
Multiple R-squared:  [redacted]  Adjusted R-squared:  [redacted]
F-statistic: [redacted],  p-value: [redacted]
```

Test the significance of the coefficients at $\alpha = 0.05$. Present all your calculations step by step.

(8 marks)

Question 1c

Evaluate whether the use of SLR model is appropriate based on the gardener's data collection method. Consider any potential violations of the assumptions resulting from the data collection method.

(8 marks)

Question 1d

Explain what adjusted R^2 is and discuss whether it differs from the normal R^2 (Multiple R-squared in R) in the context of SLR. Additionally, clarify if we should have a preference of one over the other to assess the fit of the model in this context.

(8 marks)

Question 2

Consider the following equations

$$Y_i = \alpha_0 + \alpha_1 V_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0,1),$$
$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \varepsilon_i \sim \mathcal{N}(0,1),$$

where $V_i = X_i - \bar{X}$.

Question 2a

Prove that $\hat{\alpha}_1 = \hat{\beta}_1$, where $\hat{\alpha}_1$ and $\hat{\beta}_1$ are least squares estimators.

(6 marks)

Question 2b

Prove that $\hat{\alpha}_0 = \hat{\beta}_0 + \hat{\alpha}_1 \bar{X}$.

(3 marks)

Question 2c

For the model $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$, the variances of the least squares estimates of the SLR coefficients are given by:

$$Var(\hat{\beta}_0) = s^2 \frac{\sum_{i=1}^{n} X_i^2}{nS_{xx}}, Var(\hat{\beta}_1) = s^2 \frac{1}{S_{xx}},$$

where s^2 is the unbiased estimate of $Var(\varepsilon_i)$. It is known that s^2 remains unchanged when X_i is replaced by V_i in the model $Y_i = \alpha_0 + \alpha_1 V_i + \varepsilon_i$. Discuss how $Var(\hat{\beta}_0)$ and $Var(\hat{\beta}_1)$ differs from $Var(\hat{\alpha}_0)$ and $Var(\hat{\alpha}_1)$. Justify your answer.

(4 marks)

Question 2d

Given the model:

$$Y_i = \gamma_1 \left(X_i - \bar{X}\right) + \varepsilon_i, \varepsilon_i \sim N(0,1).$$

Compare the estimated value of γ_1 in this model without an intercept, against the estimated value of α_1 in the model with an intercept and discuss any differences you observe.

(7 marks)

Question 3

Customer churn estimation is a critical aspect of business analytics and decision-making in various industries, including telecommunications, retail, and subscription services. It involves predicting whether a customer will discontinue their relationship with a company, which can have significant financial implications. The goal of churn estimation is to identify the factors and variables that influence customer attrition and to develop predictive models that can help businesses take proactive measures to retain valuable customers.

For an analysis involving a mobile subscription provider, our objective is to understand the factors influencing the probability of customer churn, measured as a percentage. We have gathered a dataset that includes the following data:

Y_i	: Churn Probability
$X_{1,i}$	: Contract Length (Months)
$X_{2,i}$	: Actual Monthly Usage (GB)
$X_{3,i}$	: Customer Satisfaction (Categorical: Low, Low/Mid, Mid/High, High)
$X_{4,i}$	: Customer Tenure, representing how long the customer has been with the provider in current contract (Months)

The objective is to explore how these "x" variables impact Churn Probability using Regression Analysis.

Question 3a

Without doing any regression analysis yet, purely based on logical reasoning, explain how do you think $X_{3,i}$, Customer Satisfaction, will affect Y_i, the Churn Probability? Justify your reasoning concisely. (Do not exceed 100 words in your answer)

(6 marks)

Question 3b

Explain how regression analysis can be used to analyze how $X_{3,i}$, Customer Satisfaction, as a categorical variable affects Y_i, the Churn Probability.

Propose a model that you can use, with reasons on why you picked that model. Also, make an educated guess on the expected signs and the expected magnitudes of the coefficients with reasoning.

(8 marks)

Question 3c

Can we use $X_{3,i}$, Customer Satisfaction, as a continuous variable instead of the categorical variable? Address the associated challenges and advantages of this choice. Also, specify what are conditions which are favourable to the use of Customer Satisfaction as a continuous variable.

(8 marks)

Question 3d

Utilizing **ALL** the available variables, outline how you would create a model to predict the probability of churn. Describe a plausible approach to remove redundant variables to make the model simpler without losing much performance.

(8 marks)

Question 4

Following up from Question 3, we wish to study the factors which will affect churn. 35 customers were recruited and the following data was collected:

Y_i	: Churn Probability
$X_{1,i}$	: Contract Length (Months)
$X_{2,i}$	: Cumulative Data Usage since start of contract (GB)
$X_{3,i}$	: Customer Satisfaction (Categorical: Low, Low/Mid, Mid/High, High)
$X_{4,i}$	: Customer Tenure, representing how long the customer has been with the provider in current contract (Months)

The summary statistics of the data is shown in Figure Q4.

```
ContractLength    Usage(GB)       CustomerSatisfaction CustomerTenure   ChurnProbability
Min.   : 3.00   Min.   : 30.00   Min.   :1.000        Min.   : 2.000   Min.   :0.01000
1st Qu.: 8.00   1st Qu.: 62.50   1st Qu.:3.000        1st Qu.: 5.500   1st Qu.:0.02500
Median :12.00   Median : 80.00   Median :3.000        Median : 9.000   Median :0.04000
Mean   :11.94   Mean   : 81.57   Mean   :3.314        Mean   : 9.057   Mean   :0.04343
3rd Qu.:14.50   3rd Qu.:102.50   3rd Qu.:4.000        3rd Qu.:11.500   3rd Qu.:0.06000
Max.   :24.00   Max.   :150.00   Max.   :5.000        Max.   :18.000   Max.   :0.10000
```

Figure Q4

Question 4a

Even before actual data work, we can already anticipate some problems with this dataset for a regression model. Write down an R script that will compute the Variance Inflation Factor (VIF) from the dataset. Explain the use of the VIF in verifying whether the assumptions are satisfied.

(7 marks)

Question 4b

Analyze the R script provided in Figure Q4b. Describe the purpose of the script and write down the regression model in the script.

```
churn_data <- churn_data %>% mutate(dur = CustomerTenure/ContractLength) %>%
                             mutate(usage_month=`Usage(GB)`/CustomerTenure)
mod2 <- lm(ChurnProbability*100 ~ ContractLength+dur+usage_month , data=churn_data )
summary(mod2)
```

Figure Q4b: R script for student's evaluation

(4 marks)

Question 4c

Compute the 95% Confidence Intervals for the coefficients of X_1 and X_2 below based on the lm summary below:

	Coef	Std err	t-value	P-val
Intercept	-7.13609	1.69023	A	<0.001
X_1 (ContractLength)	0.455	0.032	B	<0.001
X_2 (dur)	5.555	1.038	C	<0.001
X_3 (usage_month)	0.184	0.083	2.212	D

Residual standard error: 0.7721 on 31 df

Table Q4c: lm summary for multiple regression model

(6 marks)

Question 4d

If we consider adding an interaction variable X_2X_3 to the model, interpret the coefficient for this new variable within the context of the question. Refer to the provided R output in Figure Q4d for more context.

```
Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       [redacted]
ContractLength    [redacted]
dur               [redacted]
usage_month       [redacted]
dur:usage_month   0.35822    0.37557    0.954    0.348
```

Figure Q4d: R output for model with interaction term

(3 marks)

----- END OF PAPER -----